Yanxiao Liu

Homepage: https://yanxiaoliu-mike.github.io/
y.liu2@imperial.ac.uk

Background

Imperial College London **2025.10 –**
Postdoctoral Research Associate, Project: [Machine Learning and Information Theory]
Advisor: Prof. Deniz Gündüz (IEEE Fellow)

The Chinese University of Hong Kong **2021.8 – 2025.7**
PhD, Department of Information Engineering
Advisors: Prof. Cheuk Ting Li and Prof. Raymond W. Yeung (IEEE Fellow, Shannon Award 2022)

Stanford University **2024.1 – 2024.6**
Visiting Researcher, Department of Electrical Engineering
Advisor: Prof. Ayfer Özgür

The Chinese University of Hong Kong, Shenzhen **2017.9 – 2021.6**
B.Eng. in Electronic Information Engineering and Minor in Philosophy

Research

I am trained as an information theorist, and my current research bridges information theory and artificial intelligence, with the goal of making AI systems trustworthy, reliable, and scalable:

- **Information Theory**: I study nonasymptotic limits of information-theoretic problems through the lens of novel random codes. I make them practical (via low-complexity algorithms) and scalable (via network unification scheme), and investigate the fundamental limits.
  Trustworthy AI: I apply information-theoretic sampling schemes to the field of AI. I study:
  - **LLM decoding and watermarking**: I apply information-theoretic sampling algorithms to make LLM decoding faster and more reliable, and embed invisible watermarks for generative content provenance. I design novel speculative decoding algorithms for LLMs to make decoding faster and more stable, without sacrificing quality for watermarking.
  - **Algorithmic Properties of Learning**: I abstract algorithms as *channels* and use information-theoretic tools to study their fundamental limits, including compression-privacy-utility trade-offs, generalization, and memorization, and design novel algorithms to achieve these limits.
  - **Inference-time Alignment**: I develop novel best-of-n sampling schemes for LLM alignment, making LLM content safer, reducing computational costs, and mitigating reward hacking.

Selected Awards

- Chinese Institute of Information Theory Annual Conference 2025, **Best Paper Award**
- CUHK PhD International Mobility for Partnerships and Collaborations Award 2024 ($\approx 1.5\%$).
- NeurIPS Scholar Award (2024).
- ISIT Travel Award (2024).

Publications

Preprints

[1] **(α-β)** Mian Huang, **Yanxiao Liu**, and Yi Liu. "Sub-optimality of Marton's Inner Bound for the Two-Receiver Broadcast Channel". In: *https://arxiv.org/pdf/2608.19869* (2026).

[2] **Yanxiao Liu** and Mian Huang. "Counterexamples to the Markovity Conjecture for the Two-Receiver Broadcast Channel". In: *https://arxiv.org/pdf/2608.13170* (2026).

[3] **Yanxiao Liu**, Yijun Fan, and Deniz Gündüz. "Tighter Information-Theoretic Generalization Bounds via a Novel Class of Change of Measure Inequalities". In: *https://arxiv.org/pdf/2602.07999* (2026).

[4] **Yanxiao Liu** and Cheuk Ting Li. "One-Shot Information Hiding and Compound Wiretap Channels". In: (2026).

[5] **Yanxiao Liu**, Chun Hei Michael Shiu, Lele Wang, and Deniz Gündüz. "On the Generalization Error of Differentially Private Algorithms via Typicality". In: (2026).

Journal Articles

[6] **Yanxiao Liu**, Sepehr Heidari Advary, and Cheuk Ting Li. "Nonasymptotic Oblivious Relaying and Variable-Length Noisy Lossy Source Coding". In: *IEEE Transactions on Information Theory* 72.7 (2026), pp. 4555–4564.

[7] **Yanxiao Liu** and Cheuk Ting Li. "One-Shot Coding over General Noisy Networks". In: *IEEE Transactions on Information Theory* 71.11 (2025), pp. 8346–8357.

[8] Chih Wei Ling*, **Yanxiao Liu*(co-first author)**, and Cheuk Ting Li. "Weighted Parity-Check Codes for Channels with State and Asymmetric Channels". In: *IEEE Transactions on Information Theory* 70.8 (2024), pp. 5573–5588.

[9] Shenghao Yang, Jun Ma, and **Yanxiao Liu**. "Wireless Network Scheduling with Discrete Propagation Delays: Theorems and Algorithms". In: *IEEE Transactions on Information Theory* 70.3 (2024), pp. 1852–1875.

Conference Proceedings

[10] **Yanxiao Liu**, Wei-Ning Chen, Ayfer Özgür, and Cheuk Ting Li. "Universal Exact Compression of Differentially Private Mechanisms". In: *The Thirty-Eighth Annual Conference on Neural Information Processing Systems (**NeurIPS**)*. 2024.

[11] Gergely Flamich, Öykü Güner, **Yanxiao Liu**, and Deniz Gündüz. "Scalable Differentially Private Data Compression via Diffusion and Stochastic Codes". In: *ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling*. 2026.

[12] **Yanxiao Liu**, Shenghao Yang, and Cheuk Ting Li. "Joint Scheduling and Multiflow Maximization in Wireless Networks". In: *2026 International Zurich Seminar on Information and Communication (**IZS**)*. 2026, pp. 57–62.

[13] **Yanxiao Liu**, Chun Hei Michael Shiu, Lele Wang, and Deniz Gündüz. "On the Generalization Error of Differentially Private Algorithms via Typicality". In: *2026 IEEE International Symposium on Information Theory (**ISIT**)*. 2026.

[14] **Yanxiao Liu**, Sepehr Heidari Advary, and Cheuk Ting Li. "Nonasymptotic Oblivious Relaying and Variable-Length Noisy Lossy Source Coding". In: *2025 IEEE International Symposium on Information Theory (**ISIT**)*. 2025.

[15] **Yanxiao Liu** and Cheuk Ting Li. "One-Shot Coding over General Noisy Networks". In: *2024 IEEE International Symposium on Information Theory (**ISIT**)*. Vol. 71. 11. 2024, pp. 8346–8357.

[16] **Yanxiao Liu** and Cheuk Ting Li. "One-Shot Information Hiding". In: *2024 IEEE Information Theory Workshop (**ITW**)*. 2024.

[17] Yijun Fan, **Yanxiao Liu**, Yi Chen, Shenghao Yang, and Raymond W. Yeung. "Reliable Throughput of Generalized Collision Channel Without Synchronization". In: *2023 IEEE International Symposium on Information Theory (**ISIT**)*. 2023.

[18] Yijun Fan, **Yanxiao Liu**, and Shenghao Yang. "Continuity of Link Scheduling Rate Region for Wireless Networks with Propagation Delays". In: *2022 IEEE International Symposium on Information Theory (**ISIT**)*. 2022.

[19] Chih Wei Ling, **Yanxiao Liu**, and Cheuk Ting Li. "Weighted Parity-Check Codes for Channels with State and Asymmetric Channels". In: *2022 IEEE International Symposium on Information Theory (**ISIT**)*. 2022.

[20] Jun Ma, **Yanxiao Liu**, and Shenghao Yang. "Rate Region of Scheduling a Wireless Network with Discrete Propagation Delays". In: *IEEE INFOCOM 2021-IEEE Conference on Computer Communications (**INFOCOM**)*. 2021.

Professional Services

Award

- Top Reviewer for NeurIPS 2025 ($\approx 8\%$).

Service

- IEEE GLOBECOM 2026, TPC member.
- Information Theory Workshop 2024, Session Chair.

Reviewer

- Annual Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Machine Learning (ICML)
- Association for the Advancement of Artificial Intelligence (AAAI)
- IEEE Transactions on Information Theory (TIT)
- IEEE Transactions on Automatic Control (TAC)
- IEEE Transactions on Mobile Computing (TMC)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Journal on Selected Areas in Communications (JSAC)
- IEEE International Symposium on Information Theory (ISIT)

References

Raymond W. Yeung — **PhD Advisor**
Choh-Ming Li Professor — *The Chinese University of Hong Kong*
IEEE Fellow, Claude E. Shannon Award (2022)
whyeung@ie.cuhk.edu.hk

Cheuk Ting Li — **PhD Advisor**
Associate Professor — *The Chinese University of Hong Kong*
ctli@ie.cuhk.edu.hk

Deniz Gündüz — **Postdoc Advisor**
Professor, IEEE Fellow — *Imperial College London*
d.gunduz@imperial.ac.uk

Ayfer Özgür — **Visiting Host**
Professor — *Stanford University*
aozgur@stanford.edu

Vincent Y. F. Tan — **Thesis External Examiner**
Professor — *National University of Singapore*
vtan@nus.edu.sg

Chandra Nair — **Thesis Committee Chair**
Professor, IEEE Fellow — *The Chinese University of Hong Kong*
cnair@ie.cuhk.edu.hk